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Evaluation of Different Hedging Strategies for Commodity Price Risks of Industrial Cogeneration Plants

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Abstract

In this paper, we design and evaluate eight different strategies for hedging commodity price risks of industrial cogeneration plants. Price developments are parameterized based on EEX data from 2008-2011. The probability distributions derived are used to determine the value-at-risk (VaR) of the individual strategies, which are in a final step combined in a mean-variance portfolio analysis for determining the most efficient hedging strategy. We find that the strategy adopted can have a marked influence on the remaining price risk. Quarter futures are found to be particularly well suited for reducing market price risk. In contrast, spot trading of CO₂ certificates is found to be preferable compared to forward market trading. Finally, portfolio optimization shows that a mix of various hedging strategies can further improve the profitability of a heat-based cogeneration plant.

Keywords: Commodity price risk, Cogeneration, Hedging, mean-variance portfolio optimization;

JEL Classification Nos.: D22, D81, G11, G32, Q48

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1 Introduction

In light of the cold wave in Europe in February 2012, it became evident how important it is for an uninterrupted power supply that sufficient baseload power plants are in operation. The decommissioning of eight German nuclear power plants in 2011 incurred over the last couple of weeks of that year considerable challenges, particularly in Southern Germany (Flaucher, 2012). In February 2012, here and there emergency reserves had to be called upon¹, which for the power supply sector constitutes an important element to maintain the security of supply should bottlenecks arise in the electricity supply. In order to avoid such emergency situations in the future, and at the same time to account for climate change mitigation goals, targeted expansion of energy-efficient and environmentally benign baseload power plants has to be effected.

A highly energy-efficient and therefore environmentally sound technology for the constant supply of electrical energy is that of cogeneration (combined heat and power, CHP) based on the gas and steam combined cycle (CC) (Baehr, 2005). The advantages resulting from the use of this CC-CHP technology with regard to fuel consumption and CO₂ emissions led already in 2002 to the entering into force of a German federal law for the promotion of CHP plants (Kraft-Wärme-Kopplungsgesetz; KWKG, 2011). The aim of the CHP Act and related ordinances is to promote power production with this energy-efficient technology.

If heat demand is guaranteed, for example, by installing such a plant in an industrial enterprise which requires process heat throughout the year, then capacity factors of up to 90% can be realized. At the same time, the obligation to continuously provide heat energy to an enterprise imposes risks that would not occur in the case of a normal gas-fired power plant.

The continuous operation and the resulting constant fuel demand and CO₂ certificate requirements in the case of industrial CHP plants lead to considerable market price risks when purchasing the commodities (Westner & Madlener, 2011a). Moreover, electricity is produced independently of the market prices at the electricity exchange, due to the heat-demand-driven operation of the plant². Possibly unprofitable operating conditions can therefore not completely be avoided. The main reason for the occurrence of risks is the high volatility of the price

¹ Emergency or so-called “cold reserve” plants are inoperative power plants that can go online in the case of electricity shortages, despite high operating costs.

² The type of power plant investigated in this study is mainly used for generating process heat for industrial applications. Hence the flexibility in operating the power plant is strongly restricted. For more details see section 3.1.

developments at the energy exchange. Electricity in particular, due to its non-storability³, is subject to marked price variations (Vehviläinen & Keppo, 2003). A reliable estimation of possible profits or losses is thus very difficult. In order to nevertheless deal with these risks, various approaches allow for the valuation and mitigation of risk.

In this paper, we first design eight hedging strategies that offer different possibilities for hedging the above-mentioned price risks. In a next step, the price developments of electricity, natural gas, and CO₂ certificates are analyzed and the strategies scrutinized *ex post* with regard to their profitability. In order to support the selection of future strategies on scientific grounds, price developments of the years 2008-2011 are parameterized on the basis of statistical testing, and suitable probability distributions are estimated. These distributions serve to feed a model with data, which in the framework of a Monte Carlo simulation allows the assessing of the *Value at Risk* (VaR) for the choice of an individual strategy. Finally, the strategies are combined, following Markowitz' portfolio theory, in order to determine the best strategy mixes based on the prevailing risk preference of the plant operator concerned.

The remainder of this paper is organized as follows. In section 2, we review the related literature and introduce the method applied. In section 3, the model is presented in detail and the assumptions made listed and justified. In section 4.1, a brief historical analysis of the strategies is undertaken, while in section 4.2, a Monte Carlo simulation and in section 4.3, the portfolio analysis is performed. Finally, in section 5, the results are summarized and an outlook on possible further research is provided.

2 Related literature and method adopted

In this study, we first evaluate the prices published by the European Energy Exchange (EEX) for the commodities electricity, natural gas and European Union Emission Allowances (EUAs) for a historical consideration of the years 2008-2011. Next, we introduce effective strategies that enable the hedging of the price risks occurring when operating a CC-CHP plant. These strategies comprise the purchase of natural gas and EUAs and the sale of electricity at the EEX at different delivery periods, i.e. for instance in the spot and futures markets.

³ In general, electricity is considered to be non-storable in large quantities, in contrast to other commodities such as natural gas or grain. Physically, this is not quite correct, but the very high costs arising when "storing" electricity make this assumption justifiable.

Based on this, we calculate the clean spark spread⁴ summed up over the course of a year of the respective strategies, and value the latter relative to alternative strategies. Subsequent to this valuation based on historical values, the modeling of the price developments for electricity, gas, and EUAs follows, aimed at a risk valuation of the various strategies for a future year.

Energy price modeling has been a popular subject in numerous publications. Especially for the representation of price mechanisms on the power market, a rich body of literature exists. In the course of electricity market liberalization, there followed first approaches, in which attempts were made to model the electricity price developments at the exchange with well-known capital market models (see e.g. Felder, 1996; Vehviläinen & Keppo, 2003). However, it soon became apparent that the price formation of electricity due to non-storability⁵ follows other rules. Weron et al. (2001) and Guthrie & Videbeck (2007) find that simple finance models are insufficient to model the high volatilities of the spot market prices of electricity.

Modified models have thus been applied to model the specific characteristics of the electricity price development at the energy exchanges, e.g. by Escribano et al. (2002). They take various factors into account for the analysis of the spot market prices, such as seasonality and GARCH behavior (see also Duan, 1995) or time-dependent jumps, which were analyzed for their relevance on the basis of the data of different markets. Boubonnais & Méritet (2006) consider a *univariate time series approach* and Huisman et al. (2007) develop and apply a model for the representation of hourly electricity prices. Models for the price formation on the futures market were proposed and investigated, among others, by Deng (2000), Burger et al. (2004) and Bauwens et al. (2011).

For the price developments on the EUA market, albeit to a lesser extent, there also exist a number of publications and models. Benz & Trück (2009), for instance, investigate the usefulness of a *Markov-switching model* and an *AR-GARCH model* for representing the certificate price development on the European emissions trading system (EU-ETS). In contrast, El Hedi Arouri et. al. (2012) investigated the relationship between the prices for CO₂ certificates on the spot and on the futures markets by means of a *vector autoregressive (VAR) model* and a *switching transition regression-exponential GARCH model (STR-GARCH)*. Recent studies, e.g. by Chevallier (2011), deal with the instability of the volatility of EUA prices when using an *EGARCH model*.

⁴ The clean spark spread is computed as the difference between the revenues gained from electricity sales and the expenditures that result from purchasing fuel and CO₂ certificates for the amount of electricity produced, cf. eq. (1) (see also Alberola et al., 2008).

⁵ This particularity of the commodity “electricity” leads to a situation where on the markets there may be short-term scarcity and overflows that cause the prices either to rise or fall markedly. In reality, both price rises of several hundred percent and negative prices could be observed up to now.

Apart from these studies, which deal with the forecasting of EUA prices, other studies focusing on the price development of natural gas exist as well. Price volatility of gas products was already modeled, for example, by Herbert (1995). Further studies were conducted, e.g., by Pindyck (2003) or Geman & Ohana (2009). A detailed analysis of various models for the simultaneous modeling of gas and electricity spot market prices for a gas-fired power plant is provided by Heydari & Siddiqui (2010). They conclude that price volatility forecasts from non-linear stochastic models are the most accurate ones.

Based on these insights, in our study we also pursue the approach of representing the volatility by means of stochastic parameters and simulations.

In contrast to the models published so far, price developments of all commodities (electricity, natural gas, and EUAs) that are needed to compute a clean spark spread of a CC-CHP plant, are stochastically analyzed and evaluated by means of a *goodness-of-fit* test. The combined analysis of these commodities in the context of the special characteristics⁶ of a heat-demand-operated CC-CHP plant is an original contribution. To our knowledge, only Rong & Lahdelma (2007), with a focus on the planning of the CO₂ certificate trading for cogeneration plants, and Masaaki et al. (2007) have a comparable aim in their research focus.

In order to generate a database that is as broad as possible for estimating the probability distributions, we have used price series of the various commodities published by the EEX, and delivery periods over three years. First, these are level-adjusted, after which exogenous structural breaks are corrected by means of an OLS regression⁷. Subsequently, the data is analyzed by means of a selected statistical test. For energy market commodities, in particular, specific characteristics have to be taken into account. Rachev (2005) discusses for the case of the electricity price distribution the issue of extreme price jumps, and also that the kurtosis takes a relatively high value. In order to take account of these two criteria, we use two suitable test procedures for analyzing electricity prices. A standard goodness-of-fit test procedure is the *Kolmogorov-Smirnov* test. Massey (1951) showed that this test dominates other tests in the case of data with extreme price spikes. The high values of the kurtosis, which is also referred to as *fat* or *heavy tails*, and which is characteristic of electricity prices, were investigated in detail by Huisman & Huurman (2003) and Weron (2005), respectively. Weron (2005) in this context showed, in addition, that the Anderson-Darling test is especially well suited for the analysis of *heavy tails*.

⁶ Because the plant is operated based on heat demand, it cannot, for instance, be switched off arbitrarily when electricity prices are low.

⁷ Ordinary least squares (OLS) regression, see e.g. Urban & Mayerl (2011).

The price development of the other commodities (natural gas, CO₂ certificates) shows comparable structures, so that we use the Anderson-Darling method for determining a suitable underlying distribution. As a check and for the data analysis in the case of particularly imprecise values obtained with the Anderson-Darling method, we also conducted a *goodness-of-fit* test based on the *Kolmogorov-Smirnov* test procedure.

The goal of our study is to parameterize the volatility of the price development of the various commodities and products in line with Burger et al. (2004), in order to develop models for a Monte Carlo simulation that builds on this. The variation between the different models arises, because all commodities (electricity, natural gas, CO₂ certificates), based on their products on the spot and futures markets, are combined with each other, in order to analyze different hedging strategies. The models that are based on these strategies are then used for computing the clean spark spread of a 100 MW_{el} CC-CHP plant. The probability distributions thus computed for the annual clean spark spread of the various hedging strategies are, then, following the work of Henney & Keers (1999) and Lui & Wu (2008), contrasted on the basis of a *value-at-risk* (*VaR*) assessment and valued with regard to their risk. For an introduction to the *VaR* concept see, e.g., Linsmeier & Pearson (2000).

Following the *VaR* analyses, we also pursue a portfolio analysis that is based on the generated distributions of the strategy-dependent spreads, in order to determine the efficient diversification of the strategies. The approach followed is the one introduced by Markowitz (1952); see also (Markowitz, 2008). Originally developed for application to equity share portfolios, this approach has also increasingly been used for the valuation of non-finance portfolios. For instance, Westner & Madlener (2010, 2011b) show that portfolio theory can yield important new insights into the valuation of investments in power generation assets.

Our study can also be contrasted with published work due to the technical specificity of the considered heat-demand-driven power plant. Normally, the risk management of power plants is concentrated on price risks that arise when selling the electricity produced by the power plant. Nakamura et al. (2006), for example, investigated strategies for the management of price risks that are based on the optimal proportion of electricity demand and supply. Since the rated power output of the power plant considered here can only be reduced by about 30%, the option to switch off the plant in case of a negative spread is lacking. A diminution of the losses can thus only be achieved by reducing the rated power of the plant. The relationships from which the described technical particularities arise and the database used, will be discussed in more detail in the following section when the model is described.

3 The model

The decision to invest in a power plant is decisively dependent on the expected revenues from the future operation of the power plant. In order to be able to assess the level of price risk arising at the energy exchange, the *VaR* assessment is a suitable tool for supporting the decision-making. In order to apply the *VaR* approach, however, we first have to develop a model that enables the representation with sufficient accuracy of the risk to be investigated. To this end, we first explain the technical framework conditions, followed by a description of the data used and explanations on how these are prepared for the subsequent stochastic evaluation. Section 3 concludes with an introduction of the strategies used for mitigating the price risks.

3.1 Technical aspects

The conversion plant scrutinized in our investigation is a fictitious plant. In reality, CC-CHP plants are nowadays used increasingly in manifold industrial branches for providing process steam. Apart from steam production the CHP plant is also able to produce electricity. A limitation for the operation of the heat-demand-driven CHP plant is the specification of a predefined amount of process steam for the subsequent industrial processes. Without the provision of this amount of steam the subsequent process cannot be (sufficiently) sustained. Despite these restrictions in the case of a heat-driven operation, the coupling of electrical and thermal energy is a resource-saving and at the same time economically attractive variant of energy conversion, in comparison with the sole provision of process heat in a boiler.

Exemplarily, in the present study, we consider a plant with a gas turbine as the main aggregate. We assume that the gas turbine converts a certain through flow volume of natural gas under full load into about 80 MW of electrical and about 125 MW of thermal power output. The thermal energy is used for producing process steam in a condensing boiler. The steam produced is then used, on the one hand, for providing the subsequent industrial process with heat and, on the other hand, in order to additionally produce 20 MW of electricity in a downstream steam turbine. In order to ensure that the heat-using industrial enterprise receives a constant amount of process steam, an additional gas-fuelled auxiliary firing through the hot exhaust of the gas turbine is operated. The amount of gas required for the additional firing is dependent on the power level on which the gas turbine is operated. If the gas turbine is operated under full load, then the additional firing only has to provide 10 MW of thermal power.

In contrast, if it is reduced to the minimum of 70% of electrical output, the steam generator requires an additional thermal power of about 50 MW. As the absolute amount of natural gas is reduced, depending on the variation of power production in the gas turbines, this offers the possibility to react to situations with low/negative spreads. The gains or losses depend on the marginal cost by computing the difference between the electricity sold and the purchased gas and CO₂ certificates.

The operation of a power plant is profitable⁸ in hours where the revenues from selling the electricity produced exceed the cost of purchasing the amount of natural gas and CO₂ certificates required. If this is not the case, the plant produces a loss. As industrial enterprises are dependent on the process steam, the plant cannot easily be switched off. Technically, however, it is possible to lower the power output of the gas turbine and subsequently to reduce the amount of natural gas needed for a constant amount of steam production⁹. Whether or not a reduction of the power output of the gas turbine has a positive effect on plant economics depends on the price developments at the energy exchange and the plant-specific characteristics, such as turbine efficiency.

Based on conversion efficiency, plant size, the given amount of process steam, etc., it is possible to derive a relationship that allows for an assessment of the operation mode of the gas turbine. Equation (1) describes the point at which the variable marginal costs of the plant equal the marginal revenues, i.e.

$$p_{el}(E_{GT} + E_{ST}) - p_{gas} \left(\frac{E_{GT}}{\eta_{GT}} + \frac{1}{\eta_{aux}} \left(E_{th_{fix}} - \frac{E_{GT}}{\eta_{GT}} \cdot \eta_{th,GT} \right) \right) - EM \cdot p_{CO_2} \left(\frac{E_{GT}}{\eta_{GT}} + \frac{1}{\eta_{Aux}} \left(E_{th_{fix}} - \frac{E_{GT}}{\eta_{GT}} \cdot \eta_{th,GT} \right) \right) = 0, \quad (1)$$

where p_{el} denotes the electricity price in €/MWh, p_{gas} the gas price in €/MWh, p_{CO_2} the CO₂ certificate price in €/EUA, E_{GT} the electrical gross energy output of the gas turbine in MWh, E_{ST} the electrical gross energy output of the steam turbine in MWh, $E_{th_{fix}}$ the fixed, given heat load in MWh, η_{GT} the electrical efficiency of the gas turbine, $\eta_{th,GT}$ the thermal efficiency of the gas turbine, η_{aux} the efficiency of the auxiliary firing, and EM the emission fac-

⁸ When considering the “profit” at this point, only the exchange prices of the commodities are relevant. The earnings gained from selling heat, the transaction costs of trading, fixed costs of the plants, and other costs are ignored in these considerations.

⁹ In doing so, obviously, also the electrical power is reduced, so that revenues from electricity sales are reduced as well, albeit to a smaller extent.

tor for the computation of the CO₂ emissions in tCO₂/MWh. As mentioned above, due to technical restrictions the load of the gas turbine can only be reduced to 70% of the full load (see eq. (2)).

$$\begin{aligned}
 p_{el}(0.7E_{GT} + E_{ST}) - p_{gas} \left(0.7 \frac{E_{GT}}{\eta_{GT}} + \frac{1}{\eta_{aux}} \left(E_{th_{fix}} - 0.7 \frac{E_{GT}}{\eta_{GT}} \cdot \eta_{th,GT} \right) \right) \\
 - EM \cdot p_{CO_2} \left(0.7 \frac{E_{GT}}{\eta_{GT}} + \frac{1}{\eta_{Aux}} \left(E_{th_{fix}} - 0.7 \frac{E_{GT}}{\eta_{GT}} \cdot \eta_{th,GT} \right) \right) \\
 = 0,
 \end{aligned} \tag{2}$$

If we now subtract eq. (2) from eq. (1) we obtain the amount of energy that can be varied flexibly within the technical restrictions of the power plant investigated. Equation (3) illustrates the result where the constant values $E_{th_{fix}}$ and E_{ST} are eliminated and where only the financially relevant 30% load of the gas turbine is left for the further calculations.

$$\begin{aligned}
 p_{el} \cdot 0.3 \cdot E_{GT} - p_{gas} \left(0.3 \cdot E_{GT} \cdot \frac{1}{\eta_{GT}} - 0.3 \cdot E_{GT} \cdot \frac{\eta_{th,GT}}{\eta_{aux} \cdot \eta_{GT}} \right) - EM \\
 \cdot p_{CO_2} \left(0.3 \cdot E_{GT} \cdot \frac{1}{\eta_{GT}} - 0.3 \cdot E_{GT} \cdot \frac{\eta_{th,GT}}{\eta_{aux} \cdot \eta_{GT}} \right) = 0,
 \end{aligned} \tag{3}$$

In order to not only consider the equilibrium, but instead to identify the entire range in which the plant can be operated profitably, the equation is turned into an inequality. By rearranging the equation and eliminating both the factor 0.3 and E_{GT} , we obtain a simple relationship, where the dependence of the price development of all goods and the specific characteristics of the plant allow for a conclusion about the profitability in the considered time interval (see eq. (4)). If this inequality is fulfilled, i.e. if the value on the left-hand side of the equation is smaller than on the right-hand side, then a positive margin results and the plant should be operated at full load. In contrast, if the inequality is not fulfilled, a negative margin results and the power of the gas turbine should be reduced to the extent possible in order to mitigate the losses:

$$\frac{p_{el}}{p_{gas} + 0.2016 \cdot p_{CO_2}} < \frac{1}{\eta_{GT}} - \frac{\eta_{th}}{\eta_{aux} \cdot \eta_{GT}}. \tag{4}$$

Apart from these plant-specific assumptions, some further economic and technical assumptions are made for simplicity reasons¹⁰. Also, the efficiency of the plant studied is restricted to the conversion efficiencies reported here, and to related losses. Price formation for the goods themselves, however, significantly affects the profitability of the plant so that this has to be investigated in more detail.

3.2 Data used

The following evaluations and policy recommendations are based on price series for various commodities in the years 2008-2011, which are available from the EEX database. These price series differ from each other by the type of commodity (electricity, natural gas, CO₂ allowances), by delivery period, which can be traded either on the spot market or on the futures market in various time periods. In the following, the products used for the analysis are introduced by their EEX nametags, and their characteristics explained.

Spot market

Electricity:

- On the *day-ahead market*, baseload electricity contracts are traded that oblige to the physical delivery of 1 MW for 24 hours on the following day. In other words, they guarantee a constant delivery of at least 1 MW over the entire day and, therefore, cover the baseload¹¹ needs in the electricity sector. This product is referred to as “*phelix day base*” and the price is denoted in €/MWh. Price formation takes place one trading day before delivery, so that per year 365 different price notations result.
- Besides the 24 h-blocks, in our study we also consider products for electricity with an hourly resolution that oblige for the delivery of 1 MW over one hour each. These products are traded on the so-called *intra-day market* and the price is formed up to 60 minutes before delivery is due. The price is also denoted in €/MWh and the volume of the available data is thus $24 \times 365 = 8760$ ¹² observations per year (EEX, 2011a).

¹⁰ For example, we neglect costs for ramp-up and ramp-down processes that may arise. Furthermore, we assume that no transaction costs arise for the trade of goods at the energy exchange, and that control areas and delivery zones do not have any influence on price formation

¹¹ Industrial CHP plants usually operate in baseload mode, with a duration of 8000 full-load hours.

¹² This is the theoretically available value. When evaluating the data, however, it turned out that the documentation had certain gaps.

Natural gas:

- For the case of gas, we consider the prices of so-called “Natural Gas Day 1 MW Contracts”. They oblige, similarly to what we have described above, to the constant delivery of 1 MW of natural gas over the 24 hours of a specific day. In comparison to the trade of electricity, however, each natural gas contract can be traded on two days before delivery, which leads to a dataset with $2 \times 365 = 730$ observations per year. The analysis of the data has shown that the price is formed for two days, whereas trade does not take place regularly on both days. Hence, at this point, we assume that the prices that are relevant for a probability analysis are formed when trade actually takes place. This results in a data volume that is limited per year to about 400 observations. The delivery on the virtual hub takes place in the market zone of the NetConnect Germany (NCG) GmbH & Co KG and the gas traded is exclusively H-gas¹³ (EEX, 2011b).

CO₂ certificates:

- For the CO₂ certificates (EU Allowances, EUAs), we also consider spot market data. However, in contrast to the prices of electricity and natural gas, respectively, these are only determined on trading days, so that the annual data set is reduced to 250 observations. The smallest trading unit is 1 EUA, allowing the emission of one metric ton of CO₂ or CO₂-equivalents. The price is given in €/EUA (EEX, 2011c).

Futures market

Electricity:

The trading of electricity on the futures market can be done by means of different products. The price series used in this study are related to the so-called German baseload futures. These are tradable delivery periods, which determine the physical delivery of electricity supplies in the future at a specified point in time and a specified amount. Specifically, the prices of monthly, quarterly, and yearly futures are considered here, which at the EEX are named after their respective periods with their English terms:

- *German baseload month future*: The contract volume of such a delivery period is composed of the delivered power, the delivery hours, and the delivery days. For example, the total amount of delivery at 1 MW of power can be calculated as “1 MW x 30 days x 24 h/d = 720 MWh”. Depending on the number of days per month, the time period changes accordingly. The trade with such a month future starts nine months ahead at maximum

¹³ H-gas (high-gas) is natural gas with a higher energy density. In Germany, also L-gas (low-gas) is traded, which has a lower energy contents due to the addition of nitrogen (Baehr, 2005).

(cf. EEX electricity product brochure). The data analyzed for the years 2008-2011, however, show that the trade for a month future took place only six months ahead of the start of the delivery date, so that the price development investigated for this product could only be observed over six months.

- *German baseload quarter future*: This product is analogous to the month futures, with the difference that the delivery period is now three months. Also in this case, the specifications in the EEX product description deviate from the actually downloadable data. The specification indicates that the next full eleven quarters are tradable in the future, whereas the price series are only available for the next full eight quarters. In order to establish a closer link to the price developments of the years 2008-2011, we only consider at maximum the price development a year before delivery is due. For example, the price development of a quarter future for the months January, February, and March in 2009 is observed from the first trading day in 2008 up to the delivery date of the futures.
- *German baseload year future*: Also for this product, the same basic principles apply as for the already described futures contracts; in this case, the next full six years are tradable, but observable is also in this case only the price formation over a year before the delivery is due (EEX, 2011a).

Natural gas:

The delivery periods considered here for natural gas from the NCG market area are analogous to the above-described futures contracts for electricity. Specifically, these are:

- *NCG natural gas month future*: Just as in the above example, the contract size in this case is 720 MWh for a month with 30 days. These contracts are tradable for the next six months in the future and the minimal order quantity is 10 MW. The price is specified in €/MWh, with two decimals, which is the same as for the other products.
- *NCG natural gas quarter future*: comprises a contract volume of 2184 MWh if the quarter concerned can be based on 91 delivery days. Tradable are the following seven quarters and the price is given in €/MWh. As for the electricity futures, a period of one year before delivery is due forms the data basis for these contracts.
- *NCG natural gas year future*: comprises a contract volume of 8760 MWh in the case of 365 delivery days. As for electricity, contracts are tradable for the next six years and the data base is again a period of one year before delivery commences.

CO₂ certificates:

- Also the EU Emission Allowances are traded on the futures market. Liquidity, however, is lower in comparison to electricity and natural gas. This leads to a situation where only annual contracts are offered, which can be traded in each year up to the last trading day in November. These contracts, which are comparable with the year futures for electricity and gas, are, for all years, tradable in the second trading period determined after the National Allocation Plan II (NAP II)¹⁴. The smallest contract volume is 1000 EUA and the price is given in €/EUA.

3.3 Data preparation

For the data preparation needed for the subsequent analysis, we distinguish between the data base for the historical evaluation on the one hand and for the stochastic evaluation on the other hand. In the case of the historical analysis of the data, the calculation of the hourly and daily clean spark spreads serves as the basis. To this end, the data have to be matched with each other. As described in section 3.2, this means that daily available prices for gas and electricity have to be matched with the prices for CO₂ that are formed on trading days only. In order to determine the daily spread between the electricity, gas, and CO₂ certificates on a non-trading day, the price of the last trading day before trade is interrupted during the trade-free days. This means, for example, that the prices for CO₂ on a weekend (Saturday and Sunday) are the same as the price on Friday before the weekend concerned. Holidays are treated analogously.

For strategies which compare spot market prices with those at the futures market, the price of the corresponding futures contract is held constant over its delivery period, whereas that of the good traded on the spot market varies. For the consideration of the intra-day spot market, hourly prices are used in an analogous way. The price of the commodities purchased as 24 h baseload contracts is held constant over 24 h, whereas the price for the commodities traded on the intra-day market is allowed to vary.

For the stochastic evaluation conducted in the second part of our analysis, we do not focus on the accounting of the daily prices over the year, but rather on the various price developments and the volatility of the commodities over the time period considered.

¹⁴ The European Emissions Trading System (EU ETS), introduced on January 1, 2005, is divided into three periods. The period considered in our study falls into the second period (2008-2012), for which the framework conditions are stipulated in the National Allocations Plan II (NAP II); BMU – Bundesministerium für Umwelt, Naturschutz und Reaktorsicherheit (2006).

In order to allow for some insights about the data-generating process underlying each price development of a product, the data have to be prepared in different ways. To this end, the price data obtained from the EEX of the respective commodities are first sorted by year, after which their mean is calculated. This mean value is then deducted from the actual price. By computing the differences, it is possible to represent the volatilities of the prices over several years independently of level-changing¹⁵ influences (Figure 1).

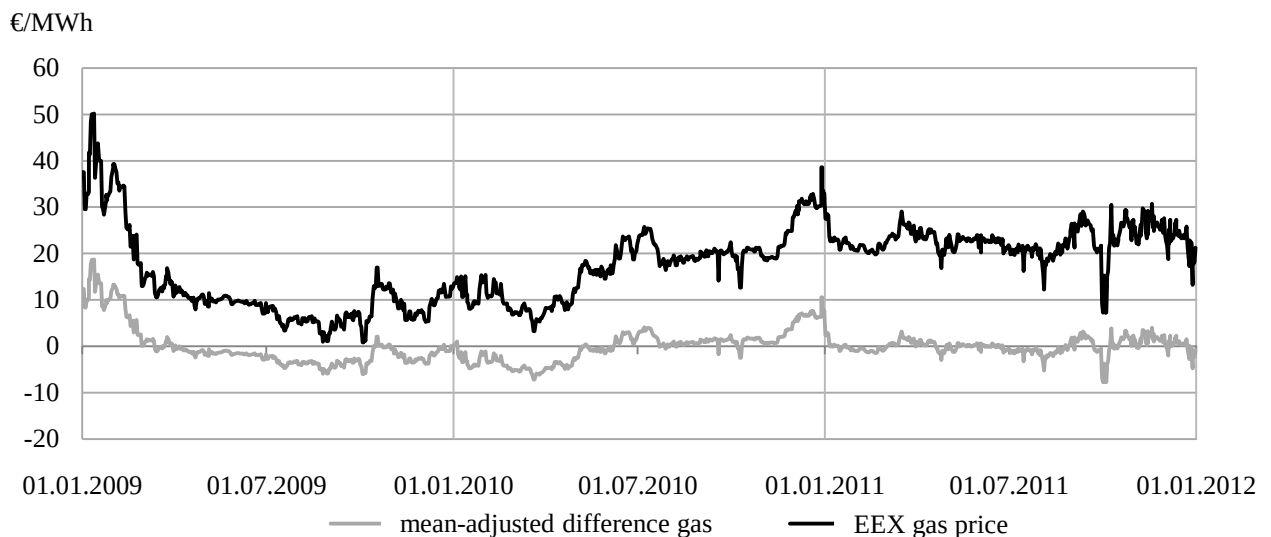


Figure 1: Price development of the EEX gas prices (lower curve: mean-adjusted difference price; upper curve: direct EEX price) on the spot market for 24 h baseload contracts over the years 2009–2011.

Source: Own calculations, based on data from www.eex.com

A further measure for the preparation of the data should reduce the impact of the financial crisis on the market price formation. For instance, for futures contracts of the commodities, a strong influence of the financial crisis on price formation can be detected for the year 2009. This is due to the fact that price formation at the exchange, depending on the product to various degrees, goes back to the year 2008. Therefore, based on the outcome of the visual inspection of the individual price trajectories of all futures contracts, some data going back to 2008 were excluded from the data analysis. For example, in Figure 2 the trajectory of the difference prices for a *German baseload quarter future* for the years 2009–2011 is shown.

¹⁵ When considering nominal prices, for example, the inflation rate or business cycle impacts (such as the growing demand for energy as a result of increased economic activity), can have a level-shifting effect.

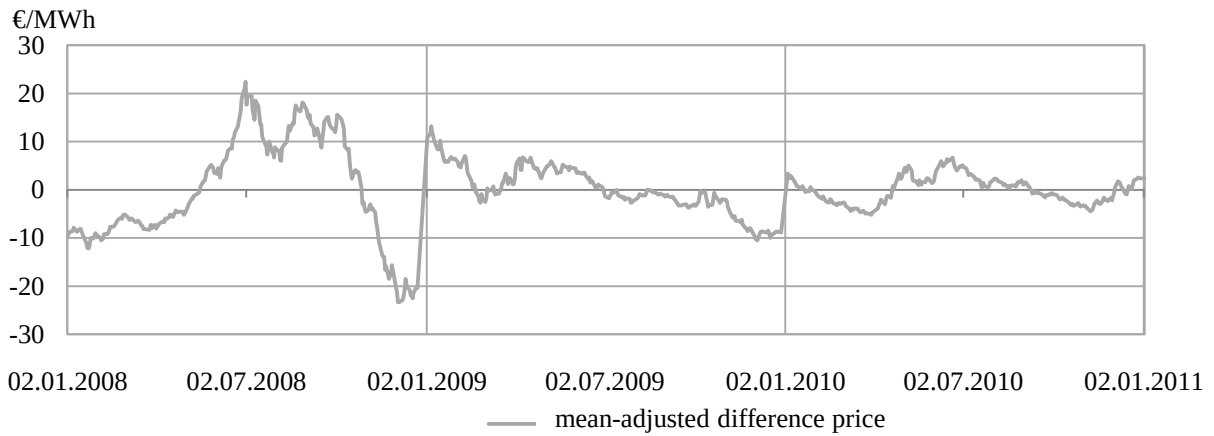


Figure 2: Mean-adjusted difference price development for baseload quarter futures (Jan, Feb, Mar) of the years 2009–2011.

Source: Own calculations, based on data from www.eex.com

As the price for the quarter future depicted in Figure 2 (January 2009) was already formed back in 2008, the influence of the financial crisis and the deviating course to the following years is particularly evident. In order to exclude this deviation from the analysis of the price series, price formation in 2008 is not considered. Therefore, as a data basis, we only use the two thirds on the right-hand side of the data depicted in Figure 2. This restriction affects the month futures for electricity and natural gas from January to May 2009, the quarter futures for electricity and gas for Q1 to Q4 in 2009, and the year futures for electricity, gas, and CO₂ certificates in 2009.

Apart from the long-term disturbances caused by the financial crisis, on March 15, 2011 the moratorium decision of the German government to shut down eight nuclear power plants, had a massive impact on the electricity price for futures contracts. Consequently, this politically induced level-shifting of the time series is treated as an exogenous structural break and depicted in Figure 3 exemplarily for an electricity month future (June 2011).

In order to account for the influence of the structural break on the data-generating process when doing the mean adjustment, we regress a constant (β_0) and a step dummy (SD_t) on the time series (F_t) and estimate these coefficients by the ordinary least squares (OLS) method:

$$F_t = \beta_0 + \beta_1 + SD_t + u_t, \quad (5)$$

where u_t is an error term and $SD_t = 0$ for $t < 15.03.2011$ and $SD_t = 1$ for $t \geq 15.03.2011$, respectively. Once the two coefficients are determined, the part of the time series before March 15 is adjusted only by the mean value (β_0), whereas the part of the time series after March 15 is adjusted by the mean value plus the effect of the exogenous shocks, ($\beta_0 + \beta_1$).

This correction method is applied to the month futures for electricity for the months April to September 2011 and for the electricity quarter futures for the 2nd, 3rd, and 4th quarters in 2011. Both for the gas price development as well as for the price of CO₂ certificates, the event on March 15, 2011, had no influence on the price formation. With the data prepared in the described manner, we can now conduct the analysis of the probability distributions of the price for each product required for the second part of the valuation.

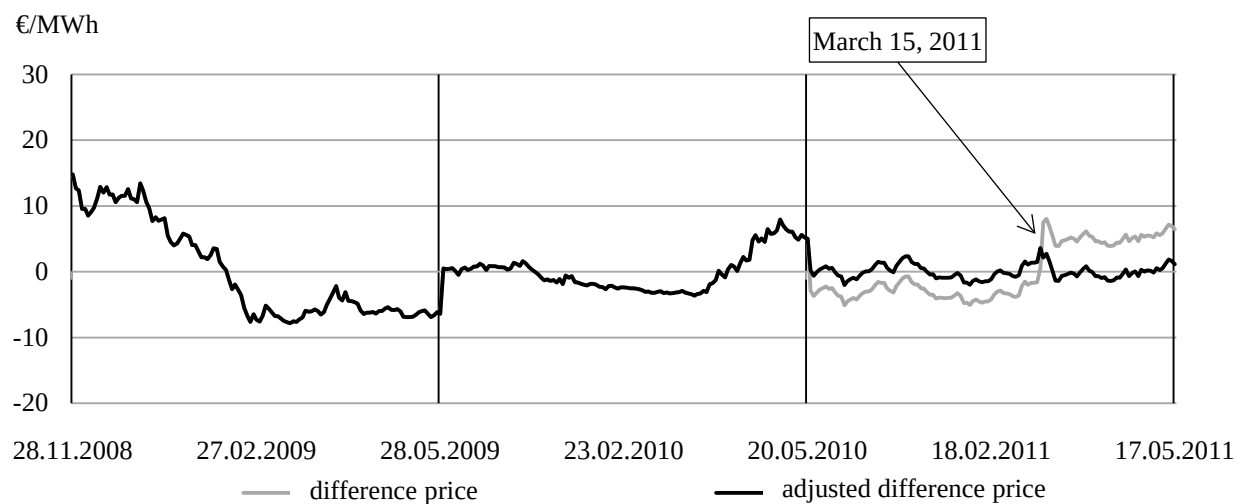


Figure 3: Difference price of the month electricity future for June and the difference price of the June future, adjusted by an exogenous structural break.

Source: Own calculations, based on data from www.eex.com

3.4 Strategy definitions

The following section defines and describes the various strategies considered in some detail. The main distinction is made according to the strategies that are based on products traded on the spot and futures market.

3.4.1 Spot market

Strategy 1 – day-ahead: This is the base case. In this strategy, all commodities are traded on the day-ahead spot market. This implies that electricity, gas, and also CO₂ certificates are traded one day before the delivery day in the form of 24 MWh blocks or corresponding amounts of certificates, respectively. The prevailing prices are geared towards the EEX daily price index.

Strategy 2 – intra-day: In this strategy, the possibility to operate the power plant flexibly within a certain range is exploited (see section 3.1). In order to react flexibly to developments of the electricity price, the amounts of electricity offered and demanded on the intra-day spot market are traded, if at all, in 1 MWh blocks. We first assume that all three commodities are bought or sold a day ahead in 24 h blocks. On the following day, the price development of electricity on the intra-day market is observed additionally. If this observation yields that the price ratio undercuts a plant-specific threshold value (see eqs. (1) to (4)), then the electrical output of the power plant is reduced. In other words, if the price of electricity on the intra-day spot market is sufficiently low, the plant owner reduces the plant's electrical output in order to save fuel and EUAs, and therefore buys electricity on the intra-day market to fulfill the delivery obligations. The advantage arises from the fact that the reduction in electrical output reduces gas consumption over-proportionally and the savings in the purchasing of gas and CO₂ certificates exceed the expenditures from electricity procurement.

3.4.2 *Futures market*

In order to investigate the price developments on the futures market and the possibilities for risk hedging by forward contracts, in the following strategies – with some exceptions for the CO₂ certificates – we only use the prices of the commodities traded on the futures market. The limitation for the futures on CO₂ certificates arises from the fact that the product range only covers annual futures, and unlike for electricity and gas, also month and quarter blocks. In order to mitigate the large, time-dependent impact of year futures, the futures contracts for electricity and gas are connected to both futures market prices (year futures) as well as to the spot market price for CO₂ certificates. The point in time at which the corresponding futures contracts are traded is a trading day that is half a period before the trading period concerned. This means, for instance, that on the 15th of the previous month, a block of electricity or gas is bought or sold for the entire following month, or that the trading day for a quarter block is chosen 1.5 months before and that for a yearly block half a year before delivery begins.

Strategy 3 – month future – CO₂ future: In this strategy, for one year, twelve monthly blocks for the commodities gas and electricity are traded, starting on December 15 of the previous year. The purchase of the corresponding year futures for CO₂ certificates is made on June 30 of the previous year.

Strategy 4 – month future – CO₂ spot: Here, as already mentioned in the introduction, electricity and gas are traded as in the strategy *month future – CO₂ future*; the difference is that CO₂ certificates in this strategy are purchased on the spot market instead.

Strategy 5 – quarter future – CO₂ future: In this strategy, the prices of electricity and gas, respectively, are determined on the basis of quarter blocks and the CO₂ price based on year blocks.

Strategy 6 – quarter future – CO₂ spot: Analogously to section 3.4.1, in this strategy, instead of year blocks for CO₂ certificates, the market price is again determined via the spot market. The prices for electricity and gas are determined analogously to the strategy *month future – CO₂ future*.

Strategy 7 – year future – CO₂ future: For this strategy, half a year in advance for each of the three commodities the entire amount for the following year is traded.

Strategy 8 – year future – CO₂ spot: In this strategy, the price of the certificates is determined on the spot market, whereas the price of the other two commodities is analyzed via the futures market through selling or buying year blocks on June 30 of the previous year.

4 Results

In this section, we first investigate, by retrospection on the basis of EEX prices for electricity, gas, and CO₂ certificates for the years 2009-2011, the profitability of the described fictitious CC-CHP plants with a total electrical output of 100 MW. The spread results from a plant-specific relationship between gas consumption and electricity production, which, depending on the plant efficiencies, can be computed by using eq. (1). The total annual spread is obtained by adding up the hourly computed spreads under the assumption that the plant is operated at full load for 8000 hours per year. For the computation of the costs arising from CO₂ emittance, a factor to compute the emitted amount of CO₂ per MWh of gas input, as published by the German Federal Environment Agency – Umweltbundesamt (UBA) and the German

Emission Trading Authority – Deutsche Emissionshandelsstelle (DEHSt)¹⁶, respectively, is used. The free allocation of certificates, as is currently practiced on the basis of the federal law Zuteilungsgesetz 2012 (BMU, 2007), is not taken into account because of the marginal cost consideration. The costs arising from the emission of CO₂ are thus totally accounted for, as they constitute a value irrespective of whether they were obtained for free (so-called “grandfathering”) or via an auction mechanism. The corresponding prices for the demanded and supplied amounts of gas, CO₂ certificates and electricity, respectively, result from the products concerned (hourly contracts, 24-hour contracts on the spot market, or month, quarter, and year contracts on the futures market) traded on the EEX or the price forecasts computed for the second part of this section.

Based on the calculated total annual spreads, the chosen strategies are analyzed with regard to their profitability / their lowest loss value and, in the following subsection 4.1, related to the global economic developments during the period studied. In subsection 4.2, then, we first describe what the model for the computation of the *VaR* looks like and which statistical distributions result, based on the *goodness-of-fit* analysis for the products concerned. After that, the results obtained from a Monte Carlo simulation are reported and analyzed. In the final subsection 4.3, we use the insights gained from the Monte Carlo simulation, in order to conduct a portfolio analysis that enables us to identify optimal strategy combinations.

4.1 Retrospective analysis of the years 2009-2011

For all strategies considered in the year 2009, the significant changes on the markets due to the financial crisis play an important role. Especially the block-wise purchase of futures contracts had an impact on 2009 that was all the stronger the more had been purchased in advance. The reason is that up to mid-2008, the prices for all commodities peaked. Therefore, the loss that arises from the purchase of year blocks is larger than the one from purchasing quarter blocks. The latter in turn is larger than the loss from purchasing month blocks. A further strong influence arises from the fact that, due to the assumptions made in the strategies *CO₂ future*, the amount of CO₂ certificates required can only be purchased in year blocks (due to the lack of subannual futures contracts). This means that the formation of prices for an annual block of CO₂ certificates took place already back in mid-2008 when the entire amount of CO₂ certificates required for 2009 were purchased. Since the market prices for CO₂ were near

¹⁶ The H-gas characteristics used here is given by UBA and the DEHSt with an emission factor of 0.056 t CO₂/GJ. This corresponds to 0.2016 t CO₂/MWh H-gas (Source: www.umweltbundesamt.de).

their peak on June 30, 2008, the influences on the profitability in the year 2009 were accordingly very distinct. This influence is put into perspective if we analyze the trade of CO₂ certificates on the spot market, as was done in the strategies *CO₂ Spot*. Table 1 provides an overview of the strategies evaluated for the year 2009.

Table 1: Accumulated spread for the years 2009-2011, different strategies.

	2009	2009	2010	2010	2011	2011
Spot market	VaR _{95%} 2009		VaR _{95%} 2010		VaR _{95%} 2011	
<i>Day-Ahead</i>	€1,713,258		€-3,205,400		€-7,642,771	
<i>Intra-Day</i>	€2,122,957		€-3,025,330		€-7,307,182	
Futures market	VaR _{95%} 2009		VaR _{95%} 2010		VaR _{95%} 2011	
	CO ₂ Future	CO ₂ Spot	CO ₂ Future	CO ₂ Spot	CO ₂ Future	CO ₂ Spot
<i>Month</i>	€-4,653,374	€1,714,623	€-987,480	€-1,199,659	€-6,320,293	€-5,318,123
<i>Quarter</i>	€-6,482,384	€-113,789	€862,878	€651,478	€-6,603,497	€-5,600,660
<i>Year</i>	€-21,302,822	€-14,933,795	€-3,314,786	€-3,527,091	€-7,255,698	€-6,253,210

Source: Own calculations

The economic advantages in 2009 that arise from the trade of all commodities in the spot market (strategies *day-ahead* und *intra-day*) are obvious. The positive balance shown results mainly from a massive decrease in the gas prices in this year. Due to the worldwide drop in the economy, the demand for natural gas on the world market dropped so much that the gas price at the exchanges was sometimes below 10 €/MWh (the maximum price for a 24 h base-load contract on September 19, 2008, was about 32 €/MWh). Although simultaneously the price for electricity in Germany also dropped, the effect in the gas prices overcompensated this decrease, so that, as a consequence, positive revenues could frequently be achieved. By trading 24 h baseload blocks on the day-ahead market, a profit of about €1.7 million could have been realized if the power plant had been operated on 8000 of 8760 h per year under full-load conditions. By operating the power plant in a variable manner, depending on the hourly intra-day market price, revenues could have been raised by about €400,000, if in each hour in which optimization was economically advantageous, backward trading was conducted (cf. Strategy 2 – *intra-day*). Note, however, that for this intra-day strategy, an additional effort not quantified here is needed for checking the market prices, as well as increased wear-and-tear of the plant. Whether or not the extra effort pays off, thus, depends largely on the arising additional costs and can, therefore, not be judged here in a final manner.

An eye-catching feature of Table 1 is that for all strategies on the futures market the block-wise purchase of certificates has a negative impact on the balance. The reason is clear

and lies in the high CO₂ certificate price on June 30, 2008 (trading day chosen for the CO₂ year futures) at the exchange. Likewise, a deteriorating balance with increasing contract length stems from the uncertainties on the market after the breakout of the financial crisis. In 2009, for futures contracts, one had to pay systematically more for trading in advance than in the spot market. This trend was already changing towards the end of 2009, and reversed in the following year 2010, when the world economy slowly picked up steam again.

Decisive for the trends in the prices of gas, electricity, and CO₂ certificates in 2010 and 2011 is the recovering global economy, in which energy is increasingly demanded again, and prices continuously rise. Table 1 (center columns) shows the results from our calculations for 2010. Due to the still dampened mood in the markets, positive annual yields can be obtained with mid-term quarter futures (strategy *quarter future* – CO₂ *future* and *spot*). If one had traded the commodities in this year month-by-month blockwise (strategy *month future* – CO₂ *future* and *spot*), in comparison to the trading with long-term contracts (strategy *year future* – CO₂ *future* and *spot*), and had traded in the spot market, the minimal loss would have occurred.

It is remarkable that by trading with intra-day products on the spot market, and by exploiting the plant flexibility as described in section 3.1, the loss in the year 2010 could have been reduced only by some €180,000. A recommendation of this strategy would, therefore, be less likely, due to the already mentioned additional expenditures.

In contrast, if one compares the results from the calculation for the year 2010 with the results obtained for the year 2009, one can see immediately that after the massive slump of the world economy (here the year 2009), short-term trading periods (spot market) tend to be in a worse position than month or quarter blocks. The expectations were still low due to the many uncertainties regarding the economic development. Still, actual trade increasingly became stronger, so that, for example the electricity prices fixed in advance were too low and the spread accordingly poor. This tendency changed as soon as the markets regained hope and the prices for longer-term contracts rose again.

The more the global economy stabilized and the more the demand returned to the level before the crisis, the worse the results are. Due to the increasing worldwide demand for gas, the price for natural gas in 2011 temporarily rose to levels close to 30 €/MWh. The resulting high fuel costs squeezed the spread for the CC-CHP plant into the negative (see Table 1, right-most columns). Interestingly, it does not seem to make much difference which strategy would have been chosen: irrespective of whether the trade was done in the spot market or in the futures market, a loss would be indispensable for all strategies. Note, however, that it was

by and large more advantageous to procure the CO₂ certificates in the spot market instead of in the futures market. The reason is that the CO₂ certificate price, contrary to the common expectation, fell strongly in 2011 (the lowest price of 6.50 €/EUA was achieved on December 14, 2011).

In the spot market, by trading in the intra-day market, and through exploitation of the technical flexibility, the loss could have been reduced by about € 330,000.

In summary, we can state that the arising negative spreads, as already discussed, are often attributable to the uncertain and high price for CO₂ certificates. If one considers the total annual yield of such a plant and the prevailing allocation rules, the CO₂ price-dependent trend will reverse. The reason for this lies in the fact that CHP plants currently still receive the more favorable allocation of emission rights according to the so-called “double benchmarking principle” (BMU, 2007). In order to promote the energy-efficient CHP technologies, they receive an allocation of certificates both for the electricity and for the heat production. Thus, in the 2nd trading period (2008-2012), this results in an excess amount of certificates. Only with the beginning of the 3rd trading period in 2013 will the additional promotion of CHP plants cease, so that profits from excess CO₂ certificates will no longer be possible.

From the insights gained from our investigations, we can conclude that the economic viability of a CC-CHP plant strongly depends on the events on the world markets¹⁷ and that by the fully accounting for the value of the CO₂ certificates (either in terms of windfall gains for the case of grandfathering, or true expenditures in the case of auctioning) the operation of such a plant does not render it economical. However, since the behavior of the markets cannot be predicted precisely, we can only try to come up with some recommendation for future decision-making, based on the existing information and experiences. In order to support this policy recommendation, in the following, we will analyze probability distributions derived in dependence of the risk measures computed.

4.2 Monte Carlo simulation and *VaR*

Building on the insights gained in the previous section, we now want to undertake a risk assessment for future decision-making. As a basis, we use the *VaR* approach already described above. In order to obtain an assessment of the percentage distribution of possible losses, we first analyze and parameterize the probability distributions of the various commodities. These

¹⁷ Of course, in this context, influences such as the weather or political changes also play a decisive role on the market price development. Such effects, however, are neglected in our analysis and left for future research.

parameterized distributions then serve as an input for the model that, as described in the previous section, is based on the same calculation mode. The only difference is that here, we do not use concrete deterministic values but instead random values that are based on the probability distributions analyzed of the individual volatilities. With the help of 100,000 runs from a Monte Carlo simulation on the price developments of the commodities, the distribution of the accumulated annual spreads is determined. Corresponding to the strategies described in section 3.4, we then compute the *VaR* for each strategy.

4.2.1 Data analysis

The data prepared according to the procedure described in section 3.3 are first investigated with a statistic *goodness-of-fit* analysis on their individual distribution. To this end, we specially take account of the shape of the volatilities on the tails of the determined distributions by using the Anderson-Darling test procedure¹⁸. Table 2 shows exemplarily¹⁹ the resulting probability distributions estimated in the analysis and the parameters for the price developments on the spot markets over the years 2009-2011.

Table 2: Estimated probability distribution and their parameter values for different price trajectories on the **spot market**, selected on the basis of test results of the Anderson-Darling (A-D) and the Kolmogorov-Smirnov (K-S) test procedures.

Commodity	Distribution	A-D	K-S	Parameter values
Electricity, day-ahead	Student <i>t</i>	1.14	0.03	Midpoint=0, Scale=7.65358, Deg. Freedom=5.16613
Gas, day-ahead	Logistic	13.89	0.07	Mean=-0.31478, Scale=1.82963
CO ₂ , day-ahead	Logistic	4.37	0.05	Mean=0.12463, Scale=1.09155
Electricity, intra-day	Logistic	64.44	0.04	Mean=-0.11086, Scale=10.92158

Source: Own calculations, based on data from www.eex.com.

Based on a histogram and the best-fitting distributions resulting from the *goodness-of-fit* test, in Figure 4 we show as an illustration the graphical representation of the analysis of the intra-day market price volatility for electricity.

¹⁸ As already described in section 2, for cross-checking, we additionally apply the Kolmogorov-Smirnov test procedure, in order to detect strongly deviating results obtained with the Anderson-Darling method, if any, and to correct them if necessary.

¹⁹ The remaining distributions analyzed are reported in Tables A1-A4 in the Appendix.

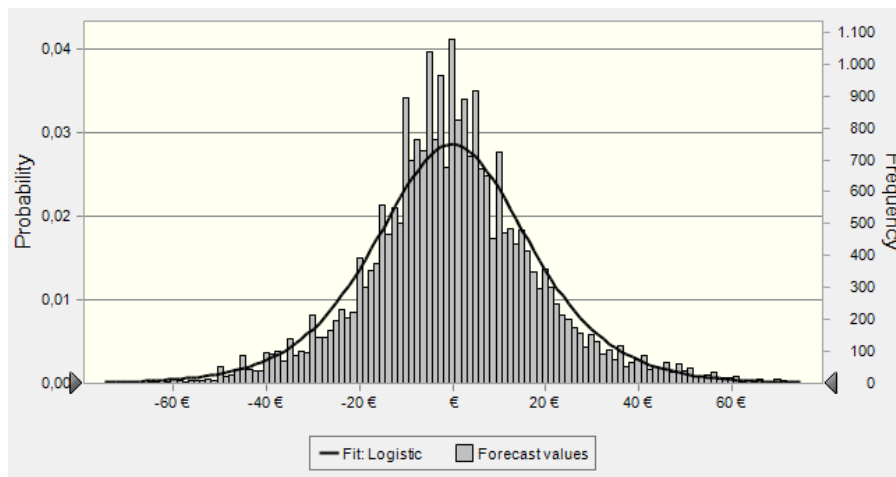


Figure 4: Histogram for the electricity price volatility on the intra-day market and graphical representation of the estimated probability distribution (logistic function).

Source: Own calculations, based on data from www.eex.com.

In order to obtain a realistic price for the year 2012 from the analysis of the distribution of the manipulated data²⁰, the price volatility has to be calculated back with an approximated mean value on the exchange price level. For that purpose, the growth rate of the data used in this study is analyzed and linearly extrapolated for the year 2012. By doing this, we deliberately ignore the striking seasonality of the prices also for the case of the futures markets by the linear extrapolation of the means. This simplification is possible owing to the fact that for the calculation of the accumulated spreads of the total year, seasonality does not play a role. Figure 5 shows how the corresponding products traded at the exchange are extrapolated on the year 2012. Especially the approximated means for the spot market prices and the year-futures of the corresponding commodities are striking. Due to the insufficient data situation, the computation of the future mean values had to be based on an analysis of a one-time and twofold price development. This restriction is tolerated here, as the strong influence of the financial crisis should not be regarded in the average growth rate anyhow²¹.

²⁰ In order to make the price development free of exogenous influences, the prices of the commodities concerned are adjusted by their annual mean values, as described in section 3.3.

²¹ This study is restricted only to the price developments after the financial crisis, since only in more recent years (incl. the year 2008) a representative price for the market of the commodities considered here was formed. Due to the splitting of the trading periods, the price development of CO₂, for example, is strongly dependent on the regulatory framework conditions. Likewise, the liquidity at the energy exchange has increased strongly over the last three years, and is not comparable anymore with the liquidity before the financial crisis.

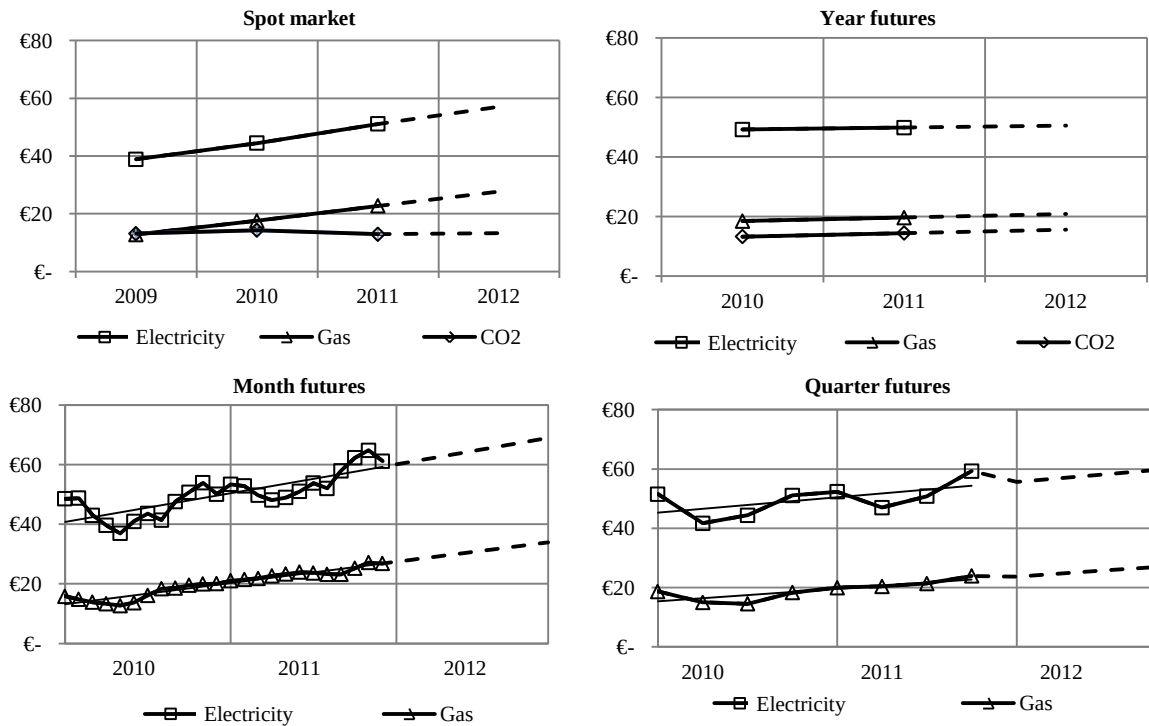


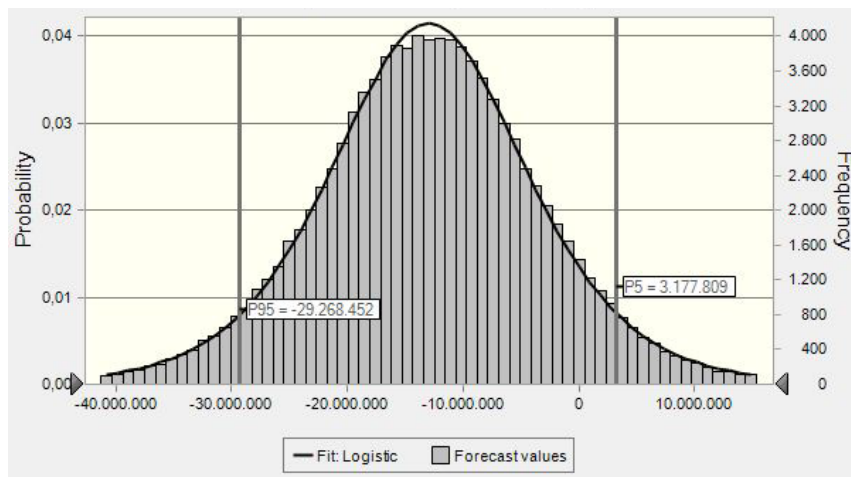
Figure 5: Average computed price of the commodities on the various markets (Spot market: top left, Futures market: top right and plots in bottom row, and their linear extrapolation for the year 2012.

Source: Own calculations, based on data from www.eex.com

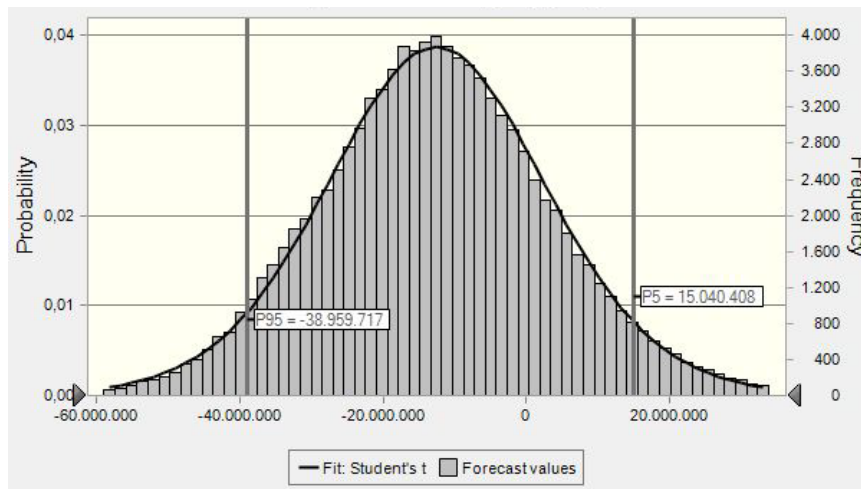
4.2.2 Monte Carlo simulation

By means of the analyzed and computed data, as described in the previous section, we can now calculate the model for the future price development based on a Monte Carlo simulation. To this end, the spreads of the corresponding strategies based on 100,000 randomly selected prices, which move within the bounds of the analyzed distributions, are computed and shown in the form of a histogram.

Figure 6a shows the probability distribution of the possible accumulated clean spark spreads for the *day-ahead* strategy. The underlying distributions used for this show a comparably high volatility due to the trade on the spot market. This leads to a situation where the entire distribution shows distinctive characteristics both in terms of positive as well as negative accumulated spreads. The $\text{VaR}_{95\%}$ is, at ca. €-28 million, the second-largest loss value computed among all scenarios considered in the analysis. This loss value, caused by the high volatility of the prices, can partly be compensated in the considerations by achieving also marked positive clean spark spreads in this strategy. This effect is reinforced for increasing volatility of the price developments considered.



(a) *Day-ahead* strategy

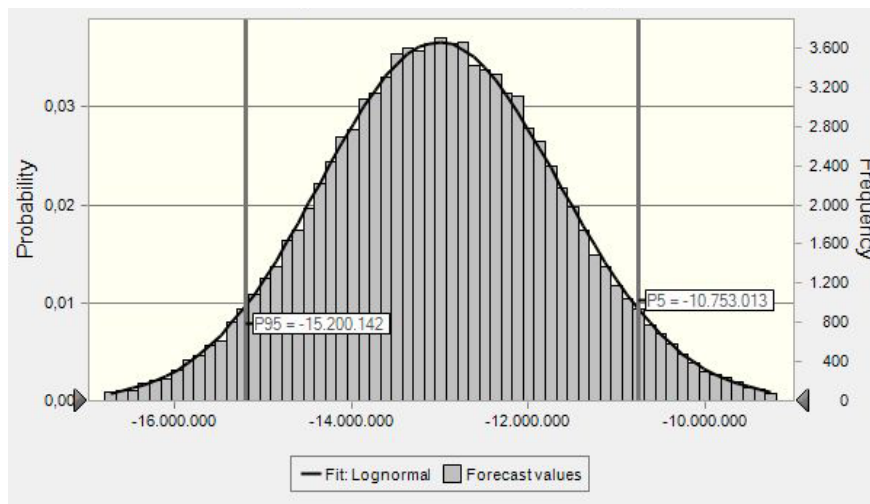


(b) *Intra-day* strategy

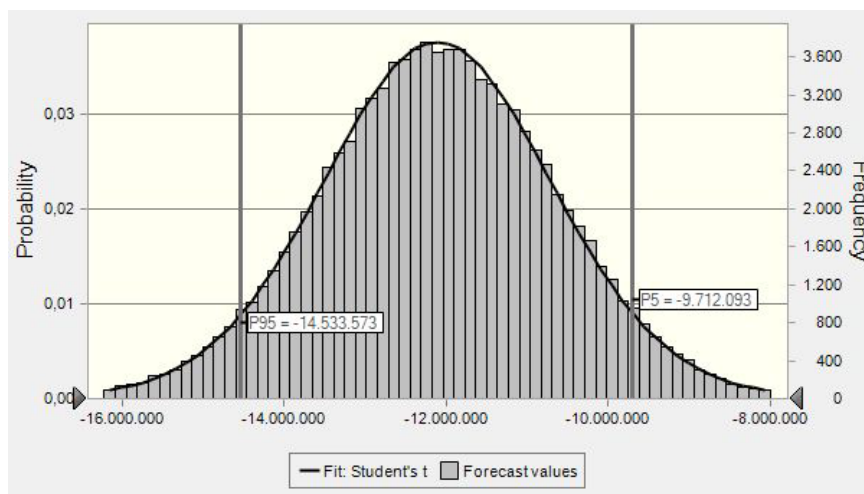
Figure 6: Simulated probability distribution of the accumulated clean spark spread for 2012 with $\text{VaR}_{95\%}$ (P95).
Source: Own calculations, based on data from von www.eex.com.

By considering Figure 6b, we can see clearly that through the pure trade of electricity on the intra-day market both the $\text{VaR}_{95\%}$ is significantly higher (ca. €-39 million), as is the probability to achieve a positive spread ($P_5\%$ ca. €15 million). However, one cannot see from these two figures that the risk of a loss incurred by the variable operation, which forms the basis of the intra-day strategy, can be reduced. This effect, however, which is visible for the historical analysis in Table 1, is overcompensated by the strongly varying intra-day market prices.

In the following analysis of the distributions, it is shown that the products traded on the future markets, compared to spot market trading, are all less risky. For the shortest delivery period (month future), we obtain significant reductions of the $\text{VaR}_{95\%}$ (see Figure 7, plots a and b). Hence for the strategies *month future – CO₂ future* and *spot* we get VaR values of about €-15 million and €-14 million, respectively.



(a) *Month future – CO₂ futures* strategy



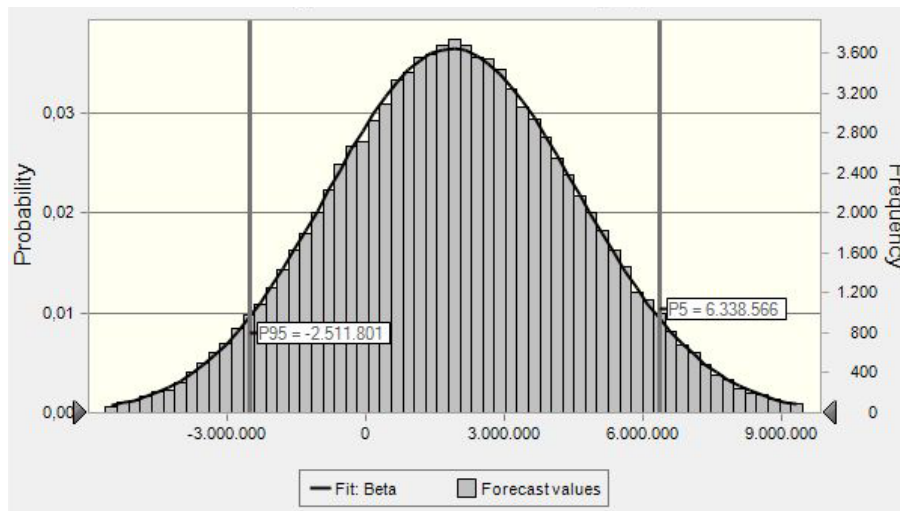
(b) *Month future – CO₂ spot* strategy

Figure 7: Simulated probability distribution of the accumulated clean spark spread for 2012 with VaR_{95%} (P95).
Source: Own calculations, based on data from www.eex.com.

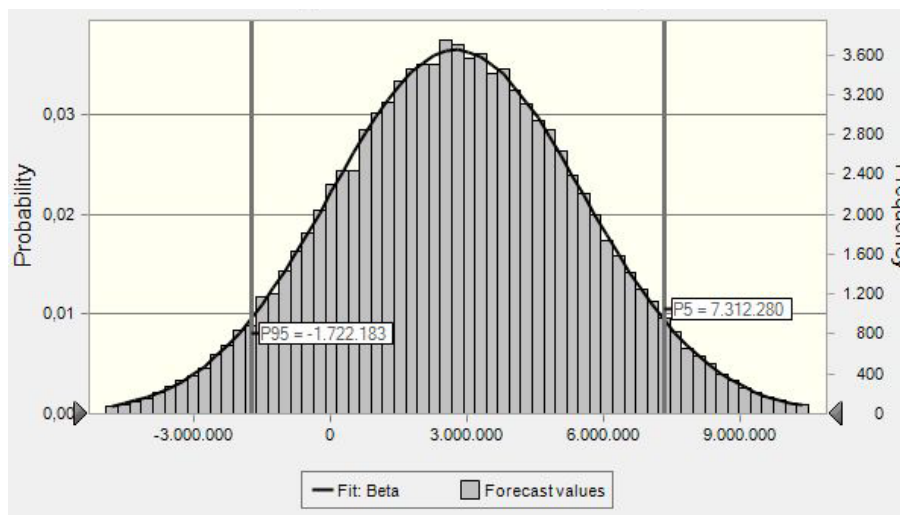
It is striking that the variance in these two strategies is lower by a factor of about 100, compared to the strategies *day-ahead* and *intra-day*. This leads, on the one hand, to a situation where the bandwidth of possible price developments is considerably lower and, on the other hand, that the best case for both strategies that occurs with a 5% probability is (or can be) negative. The reason for this lies in the lower price fluctuations of the month futures compared to the spot market considerations.

This trend is confirmed when considering the probability distributions of the strategies *quarter future – CO₂ future* and *spot* (see Figure 8, plots a and b). This span between the 95% and the 5% percentiles is lowered successively for the last of the strategies considered. For instance, the Delta between the 95% and the 5% percentiles for the analysis of the intra-day- and day-ahead market prices lies between €55–35 million, respectively. The Delta for the

consideration of the month futures now lies only at about €25 million and that of the quarter blocks at only about €8 million. The well-known weaker manifestations of the price variations from the futures market are thus well represented by the model and corroborate its viability.



(a) *Quarter future – CO₂ futures* strategy

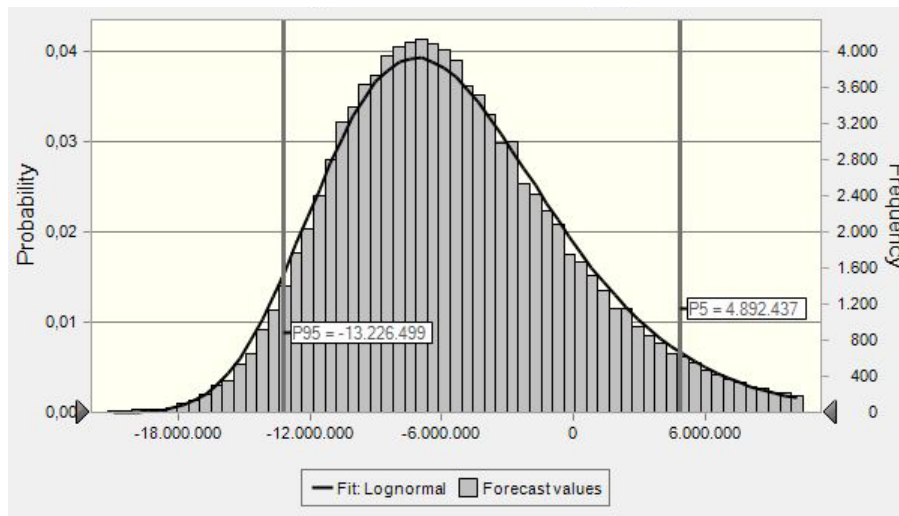


(b) *Quarter future – CO₂ spot* strategy

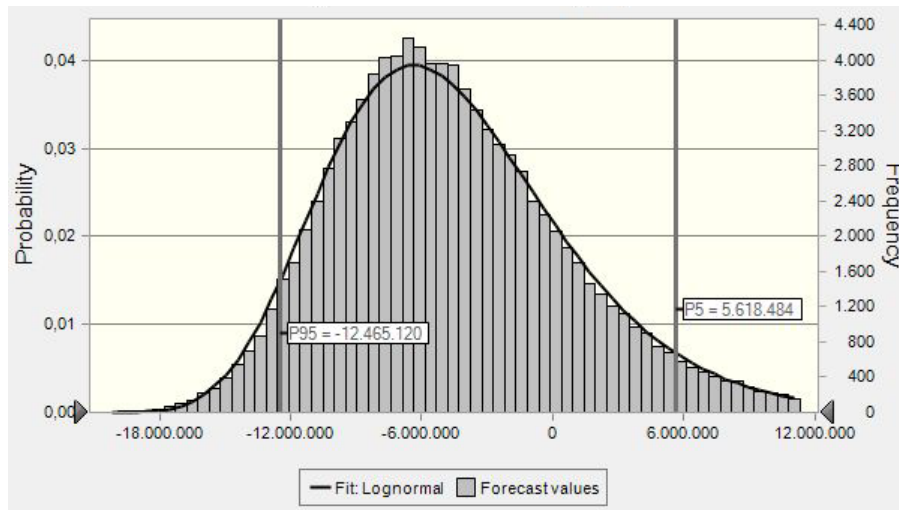
Figure 8: Simulated probability distributions of the accumulated clean spark spread for 2012 with VaR_{95%} (P95).
Source: Own calculations, based on data from www.eex.com.

Worth mentioning about the distribution of the *Quarter Future* strategies is also that the possible positive net revenues at €6 and 7 million are relatively high. From this, we can derive that the concentration in the trade of quarter blocks is a good way to reduce the loss risk and to not exclude possible gains. Possibly, the trade with quarter blocks offers a good compromise between the loss of flexibility and security, respectively.

If we increase, furthermore, the delivery period and confine trading to complete annual blocks, we get the distributions depicted in Figure 9, plots a and b.



(a) *Year Futures – CO₂ futures strategy*



(b) *Year futures – CO₂ spot strategy*

Figure 9: Simulated probability distribution of the accumulated clean spark spread for 2012 with VaR_{95%} (P95).
Source: Own calculations, based on data from www.eex.com

From this we can see that the VaR_{95%} in comparison to the quarter futures rises again, which results from the lowered flexibility. Additionally, a methodical reason for this development could be caused by the fact that the approximated average price development is relatively inaccurate due to the small sample.

All investigated strategies on the futures market have in common that the change from the futures to the spot market products in the trade of CO₂ certificates improves the results. The main reason for this lies in the representation of the trend that resulted in 2011. Until year's end 2011, the price of certificates sank from more than €15 to only about €6, which

leads to a situation that a rise of only -0.105 was assumed for the linear extrapolation of the mean value, so that the trade with spot market products is thus more advantageous. A further commonality that can be taken from the distributions is that also here for increasing risk the possible profit rises.

Table 3: Simulated VaR_{95%} 2012 for the year 2012.

Strategies for the spot market		
	VaR _{95%} 2012	
Day-ahead	€-29,268,452	
Intra-day	€-38,959,717	
Strategies for the futures market		
	VaR _{95%} 2012	
	CO ₂ future	CO ₂ spot
Month	€-15,200,142	€-14,533,573
Quarter	€-2,511,801	€-1,722,183
Year	€-13,226,499	€-12,465,120

Source: Own calculations

Table 3, finally, provides an overview of the VaR_{95%} values of the individual strategies described up to now. This table, containing the mean values and the standard deviations of the distributions analyzed, are used in the following section to determine an optimal combination of strategies in the framework of a portfolio analysis.

4.3 Combination of different strategies

In this section, based on Markowitz' portfolio theory, we investigate the diversification of the various strategies with regard to efficient mixes. Efficient in this context means that the portfolio for a given risk level cannot yield a higher clean spark spread. For the analysis after Markowitz, we first define the mean value and the corresponding standard deviation²² of each strategy, assuming a normally distributed frequency distribution (Table 4). Additionally, we determine the correlations between the strategies (Table 5).

²² The fact that the distributions of the spreads of the corresponding strategies are not always normally distributed, and that this results in an error, is known and considered in the valuation.

Table 4: Mean values (m_i) and standard deviations (σ_i) of the frequency distributions for Strategies 1–8 (in €).

Mean	Standard deviation	
m_1 -12,977,951	σ_1	10,089,526
m_2 -12,598,221	σ_2	16,553,714
m_3 -12,989,615	σ_3	1,353,674
m_4 -12,114,645	σ_4	1,466,302
m_5 1,905,039	σ_5	2,686,234
m_6 2,779,616	σ_6	2,745,150
m_7 -5,381,124	σ_7	5,649,795
m_8 -4,557,947	σ_8	5,657,007

Source: Own calculations

Table 5: Computed correlation coefficients ($\rho_{i,j}$) of Strategies 1–8.

Correlation								
$\rho_{i,j}$	$\rho_{i,1}$	$\rho_{i,2}$	$\rho_{i,3}$	$\rho_{i,4}$	$\rho_{i,5}$	$\rho_{i,6}$	$\rho_{i,7}$	$\rho_{i,8}$
$\rho_{1,j}$	1.00	0.82	0.43	0.59	0.10	0.50	-0.26	-0.24
$\rho_{2,j}$		1.00	0.41	0.51	0.12	0.42	-0.17	-0.17
$\rho_{3,j}$			1.00	0.79	0.54	0.56	0.29	0.23
$\rho_{4,j}$				1.00	0.24	0.67	-0.21	-0.17
$\rho_{5,j}$					1.00	0.62	0.71	0.68
$\rho_{6,j}$						1.00	0.02	0.09
$\rho_{7,j}$							1.00	0.94
$\rho_{8,j}$								1.00

Source: Own calculations

Based on this, using eq. (6) the mean of the portfolio m_p

$$m_p = \sum_{i=1}^n w_i \cdot m_i, \quad (6)$$

with m_i being the mean value of the distribution of the strategy i analyzed and w_i as weighting coefficient, is computed. The pre-factor w_i considers here the percentage shares of the various strategies and in sum yields, following eq. (7), always unity or 100%, i.e.:

$$m_p = \sum_{i=1}^n w_i \cdot m_i, \quad (7)$$

The variance σ_p (standard deviation) of the portfolio is computed by using the equation

$$\sigma_p = \sqrt{\sum_{i=1}^n w_i^2 \cdot \sigma_i^2 + \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n w_i w_j \cdot \sigma_i \sigma_j \rho_{i,j}}, \quad (8)$$

where $\rho_{i,j}$ is the correlation between different strategies, and σ_i and σ_j are the standard deviations of the respective strategies.

The different strategy combinations that result from the formulas described here are depicted in Figure 10.

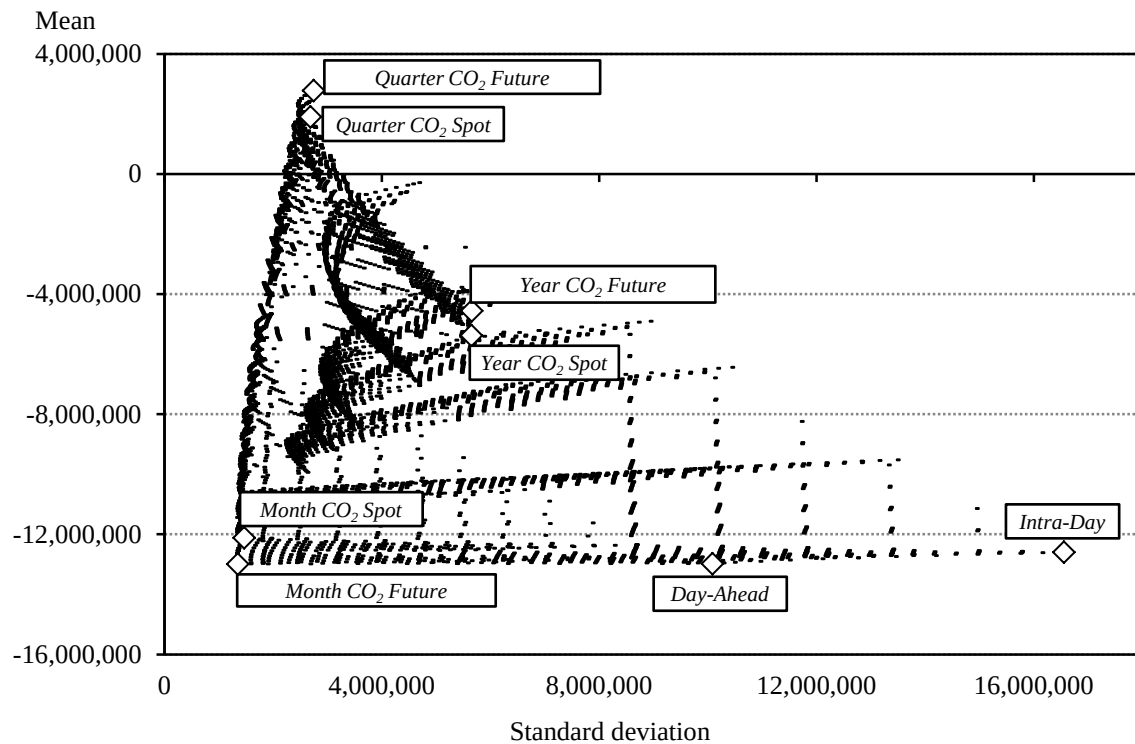


Figure 10: Portfolio analysis of the eight strategies and their positioning as single strategies.

Source: Own calculations

Each point in the diagram represents a portfolio that is composed of different shares of the individual strategies. The negative clean spark spread of most of the strategies that was already shown in the previous section is also catching the eye in this figure. Only few portfolios generate a positive spread. Hence, a main focus in these considerations is put on keeping the losses low at minimum risk or, depending on the risk preference of the decision-maker, to generate a positive clean spark spread. In this representation, the various risk distributions and expected mean values of the individual strategies can easily be seen. One can see that the two strategies pursued on the spot markets (*day-ahead* and *intra-day*), if they are the only ones

adopted, lead to a very high risk and at the same time low expected mean values. Very low mean values are also depicted for the two strategies *month CO₂ future* and *spot*, which, however, in contrast to the two spot market strategies, show a very low variance (risk). The two strategies that are based on yearly contracts, *year CO₂ future* and *spot*, in comparison to the strategies discussed so far, have higher expected mean values and thus lead to a lower average loss. In contrast to the strategies that are based on monthly contracts, this profit at the mean value results from accepting a higher risk.

As already for the case of the *VaR* analysis, here also the two strategies based on quarterly contracts show the most favorable properties. Both a positive mean value and a comparably low risk can be expected for these strategies. They can only be improved by portfolios characterized by points that lie still further to the left. These portfolios have a similar mean value but at a lower risk.

In the case of the consideration of the positive spreads with higher granularity, as is shown in Figure 11, one can see that by the equivalent combination of the two strategies that are based on quarterly contracts, the risk can be mitigated and the mean value lies between the two points of the pure strategies.

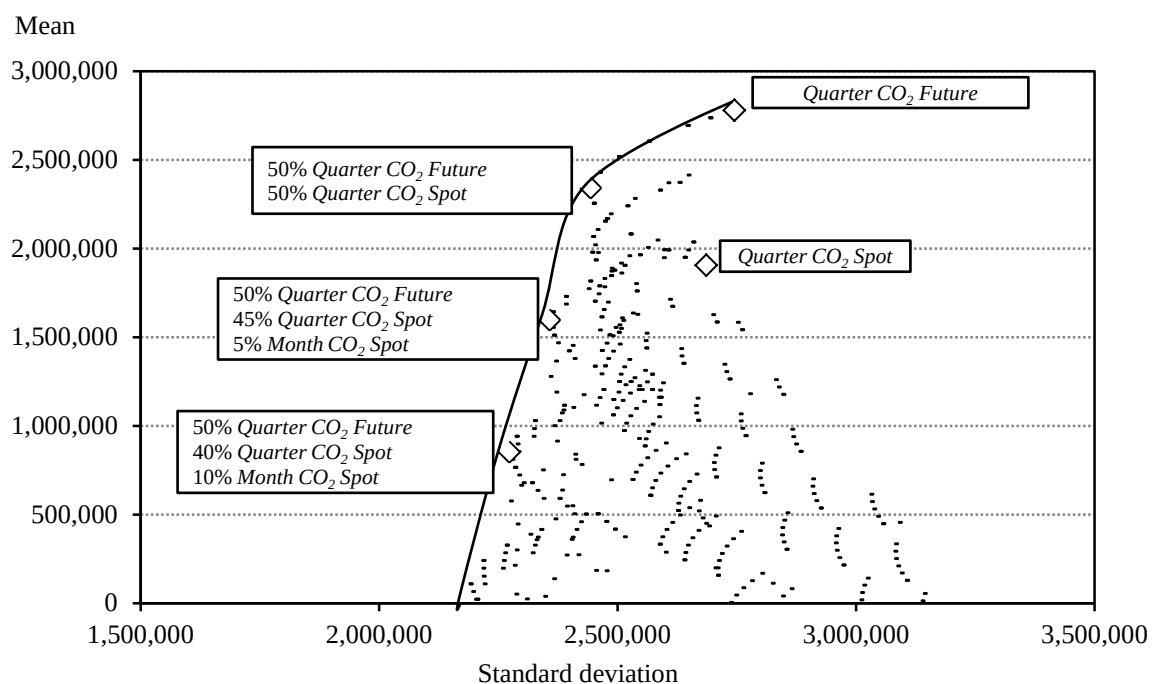


Figure 11: Excerpt of the portfolio analysis and representation of possible portfolio variants for risk mitigation.

Source: Own calculations

As can be seen, the pursuit of a single strategy ignores the possibility that by diversification of the portfolio the risk can be reduced while maintaining a constant average spread.

Figure 11, as an illustration, also depicts the shifting of the shares of the various strategies within a portfolio change according to the selected points. For instance, if a portfolio is chosen with a minimal risk, the *month CO₂ spot* strategy gains more and more in importance, whereas the influence of the *quarter CO₂ spot* strategy decreases. The level of risk that can be tolerated, and which average clean spark spread can be expected, eventually depends on the risk preference of the decision-maker.

In conclusion, we can state that by combining medium- and long-term futures contracts, the expected mean value can be raised, whereas simultaneously the risk is reduced. We also find that, by diversification of the strategies, losses can be reduced and the risk exposure diminished.

5 Summary and outlook

By hedging against market price risks, the operators of industrial CHP plants are subject to rising risks as the volatility of price developments on the commodity markets increases. The partly large risks due to very volatile commodity prices make it necessary to systematically evaluate and hedge. When operating a heat-driven industrial CHP plant, some special characteristics need to be considered.

In this study, we show that by using the technically feasible flexibilities in plant operation, economic savings can be achieved. However, due to heat-based operation and the resulting limited power reductions possible, these savings are smaller than in the case of electricity-driven power plants. Hence, it cannot be used without bounds for variable operation, as the additionally arising costs have to be taken into account and balanced against the additional revenues that accrue from the variable operation of the plant.

When valuing future operating decisions, we have shown by means of Monte Carlo simulation that the market price risk in dependence of the selected trading product and the resulting strategy can be affected drastically, and that the choice of a suitable strategy diminishes the *VaR*. More specifically, when trading short-term products on the spot market, these products feature high price volatilities, so that high loss values may result; conversely, however, there is also the possibility of reaping high gains in case commodity prices move in a favorable direction. Moreover, we could show that the market price risk can be mitigated by trade on the futures market. In particular, the trade with quarterly products was shown to be advantageous.

Fundamentally, we could show that the trade of CO₂ certificates on the futures market always leads to worse results than trade on the spot market. Due to uncertainties on the CO₂ market, and due to the still existing overflow of certificates (and resulting low prices) for all strategies, spot market trading is recommended.

Finally, we could show by means of an analysis of combined strategies borrowed from Markowitz' portfolio theory that the diversification of different strategies also for heat-driven CHP plants can lead to an improvement of the result. This enables decision-makers, depending on their risk preference, to choose the subjectively best mix of strategies.

All results presented in this paper have to be interpreted as recommendations to support decision-making that helps to mitigate market price risks. Weaknesses of the approach introduced lie in the partly small sample sizes of the data, which makes the generalization of the results difficult. In order to improve the precision of such analysis, future analyses in this field ought to be concentrated on the enlargement of the data sample for raising forecast accuracy. Moreover, in this study we have also shown the possibility to vary the different trading products amongst each other, which was only exploited to a limited extent, so that a further investigation on the interdependencies of different delivery periods is recommended. Additionally, a validation of the models specified, based on historical data, would markedly improve their explanatory power, and thus also ought to be subject to future research in this field.

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Appendix: Estimation of the probability distribution

Table A1: Estimated probability distributions and their parameter values for different price trajectories and months at the **spot market** for electricity, natural gas, and CO₂.

Commodity	Distribution	A-D	K-S	Parameters
Electricity, Day-Ahead	Student's t	1.14	0.03	Midpoint=0, Scale=7.65358, Deg. Freedom=5.16613
Gas, Day-Ahead	Logistic	13.89	0.07	Mean=-0.31478, Scale=1.82963
CO ₂ , Day-Ahead	Logistic	4.37	0.05	Mean=0.12463, Scale=1.09155
Electricity, Intra-Day	Logistic	64.44	0.04	Mean=-0.11086, Scale=10.92158

Notes: Anderson-Darling (A-D) and Kolmogorov-Smirnov (K-S) test procedures.

Source: Own calculations, based on data from www.eex.com

Table A2: Estimated probability distributions and their parameter values for different price trajectories and months at the **futures market for electricity and natural gas**.

Month	Distribution	A-D	K-S	Parameters
Electricity Month Future:				
Jan	Min Extreme	2.59	0.07	Likeliest=1.4691, Scale=2.49619
Feb	Beta	0.36	0.04	Minimum=-7.67577, Maximum=5.57878, Alpha=1.96589, Beta=1.42882
Mar	Beta	0.54	0.04	Minimum=-4.67179, Maximum=5.80386, Alpha=1.59371, Beta=1.9799
Apr	Beta	0.47	0.04	Minimum=-4.05867, Maximum=7.35096, Alpha=1.62438, Beta=2.94346
May	Max Extreme	0.39	0.03	Likeliest=-1.0285, Scale=1.82959
Jun	Lognormal	5.10	0.10	Mean=0.00178, Std. Dev.=4.61528, Location=-15.85035
Jul	Lognormal	0.71	0.05	Mean=0.01119, Std. Dev.=4.19601, Location=-8.24919
Aug	Weibull	0.77	0.05	Location=-9.39715, Scale=10.48361, Shape=3.4796
Sep	Min Extreme	1.01	0.05	Likeliest=1.53334, Scale=2.76426
Oct	Weibull	0.47	0.03	Location=-16.83904, Scale=17.85984, Shape=8.19619
Nov	Gamma	3.61	0.07	Location=-82.66186, Scale=0.08274, Shape=999
Dec	Logistic	2.44	0.06	Mean=0.13666, Scale=1.53807
Gas Month Future:				
Jan	Triangular	2.21	0.10	Minimum=-5.76667, Likeliest=-0.8071, Maximum=6.24523
Feb	Gamma	0.81	0.06	Location=-5.4896, Scale=0.73421, Shape=7.47684
Mar	Min Extreme	0.54	0.05	Likeliest=0.60089, Scale=1.07394
Apr	Lognormal	0.64	0.06	Mean=0.00197, Std. Dev.=1.16847, Location=-4.1905
May	Beta	0.35	0.04	Minimum=-2.62514, Maximum=4.4736, Alpha=2.81547, Beta=4.79795
Jun	Gamma	1.29	0.07	Location=-4.13838, Scale=1.35984, Shape=3.04328
Jul	Max Extreme	1.70	0.07	Likeliest=-0.96273, Scale=1.76253
Aug	Lognormal	0.92	0.06	Mean=0.01147, Std. Dev.=1.75404, Location=-9.72848
Sep	Weibull	0.80	0.06	Location=-5.48555, Scale=6.02296, Shape=4.53664
Oct	Logistic	2.05	0.08	Mean=-0.01864, Scale=0.85603
Nov	Lognormal	2.25	0.06	Mean=0.0005, Std. Dev.=1.80361, Location=-16.32555
Dec	Logistic	0.96	0.05	Mean=-0.0231, Scale=1.23943

Notes: Anderson-Darling (A-D) and Kolmogorov-Smirnov (K-S) test procedures

Source: Own calculations, based on data from www.eex.com

Table A3: Estimated probability distributions and their parameters for different price developments and **quarters on the electricity and gas futures market.**

Quarter	Distribution	A-D	K-S	Parameters
Electricity Quarter Future:				
Jan-Mar	Logistic	1.26	0.04	Mean=0.00239, Scale=2.18294
Apr-Jun	Weibull	0.79	0.03	Location=-12.06335, Scale=13.06121, Shape=5.52762
Jul-Sep	Weibull	2.13	0.06	Location=-13.70833, Scale=14.68613, Shape=6.72084
Oct-Dec	Logistic	1.14	0.05	Mean=0.06771, Scale=1.45254
Gas Quarter Future:				
Jan-Mar	Weibull	1.92	0.05	Location=-15.81943, Scale=17.05384, Shape=5.98798
Apr-Jun	Beta	1.21	0.06	Minimum=-5.94769, Maximum=6.82688, Alpha=2.68774, Beta=3.08504
Jul-Sep	Triangular	3.59	0.08	Minimum=-3.8928, Likeliest=-1.86846, Maximum=6.17001
Oct-Dec	Beta	1.55	0.07	Minimum=-5.21937, Maximum=4.73666, Alpha=2.42559, Beta=2.20126

Notes: Anderson-Darling (A-D) and Kolmogorov-Smirnov (K-S) test procedures.

Source: Own calculations, based on data from www.eex.com

Table A4: Estimated probability distributions and their parameters for different price developments **on the electricity, gas and CO₂ certificate futures market.**

Commodity	Distribution	A-D	K-S	Parameters
Year Future:				
Electricity	Logistic	0.75	0.03	Mean=0.00101, Scale=1.52884
Gas	Min Extreme	3.54	0.07	Likeliest=1.2134, Scale=2.10149
CO ₂	Min Extreme	1.55	0.05	Likeliest=0.60352, Scale=1.03397

Notes: Anderson-Darling (A-D) and Kolmogorov-Smirnov (K-S) test procedures.

Source: Own calculations, based on data from www.eex.com



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